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RECALL

→ Bilinear Mapping

$f: U \times W \longrightarrow V$ is bilinear if

1) $f(c_1u_1 + c_2u_2, W) = c_1 f(u_1, w) + c_2 f(u_2, w)$

2) $f(u, c_1w_1 + c_2w_2) = c_1 f(u, w_1) + c_2 f(u, w_2)$

→ Bilinear Form

bilinear mapping of type $f: U \times V \longrightarrow \mathbb{R}$ is bilinear form

i.e codomain of f is $\mathbb{R}$

Ex $f: \mathbb{R}^2 \times \mathbb{R} \longrightarrow \mathbb{R}$

$$f\left(\begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}\right) = 3u_1v_1 - 4u_2v_1 + v_2u_2$$

1) find matrix of transformation if $\mathbb{R}^2 = \text{span}\left\{\overset{e_1}{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}, \overset{e_2}{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}\right\} = B$

2) find matrix of transformation if $\mathbb{R}^2 = \text{span}\left\{\underset{e_1'}{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}, \underset{e_2'}{\begin{pmatrix} 1 \\ -1 \end{pmatrix}}\right\} = B$

1 $$F_B = \begin{pmatrix} \overset{a_{11}}{f(e_1, e_1)} & \overset{a_{12}}{f(e_1, e_2)} \\ f(e_2, e_1) & f(e_2, e_2) \end{pmatrix} = \begin{pmatrix} 3 & ? \\ ? & ? \end{pmatrix}$$

$$f(e_1, e_1) = f\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = 3(1)(1) - 4(0)(1) + (0)(0) = 3$$

2 $$F_{B'} = \begin{pmatrix} f(e_1', e_1') & f(e_1', e_2') \\ f(e_2', e_1') & f(e_2', e_2') \end{pmatrix} = \begin{pmatrix} ? & ? \\ 5 & ? \end{pmatrix}$$

$$f(e_2', e_1') = f\left(\begin{pmatrix} 1 \\ -1 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right) = 3(1)(1) - 4(-1)(1) + (-1)(2) = 5$$

The relation b/w F_B & $F_{B'}$ is $F_{B'} = \underset{B \leftarrow B'}{(P)} \; F_B \; \underset{B \leftarrow B'}{P}$

$$P_{B \leftarrow B'} = \left[[e_1']_B \quad [e_2']_B \right]$$

$$e_1' = xe_1 + ye_2$$

Dated:

$$\begin{pmatrix}1\\2\end{pmatrix} = x\begin{pmatrix}1\\0\end{pmatrix} + y\begin{pmatrix}0\\1\end{pmatrix}$$

$x = 1,\ y = 2$

- $f : U \times U \to \mathbb{R}$ where $U = \text{span}\{\overset{e_1}{\cos x}, \overset{e_2}{\sin x}, \overset{e_3}{1}\}$

$$f(u, v) = \int_{-\pi}^{\pi} uv\, dx$$

Find matrix of transformation

Sol:

$$F_B = \begin{pmatrix} \overset{a_{11}}{f(e_1, e_1)} & f(e_1, e_2) & f(e_1, e_3) \\ f(e_2, e_1) & & \\ f(e_3, e_1) & & \end{pmatrix}_{3\times 3}$$

where $f(e_1, e_3) = f(\cos x, 1) = \int_{-\pi}^{\pi} \cos x\, dx$

Definition: A bilinear form is said to be symmetric if

$$f(u, v) = f(v, u)$$

e.g:

$$f\left(\begin{pmatrix}u_1\\u_2\end{pmatrix}, \begin{pmatrix}v_1\\v_2\end{pmatrix}\right) = u_1 u_2 - v_1 v_2$$

A bilinear form is said to be asymmetric | alternating | skew symmetric | anti symmetric $(F_B^T = -F_B)$

if $f(u, v) = -f(v, u)$

e.g: $f\left(\begin{pmatrix}u_1\\u_2\end{pmatrix}, \begin{pmatrix}v_1\\v_2\end{pmatrix}\right) = u_1 u_2 - v_1 v_2$

$$f\left(\begin{pmatrix}v_1\\v_2\end{pmatrix}, \begin{pmatrix}u_1\\u_2\end{pmatrix}\right) = v_1 v_2 - u_1 u_2$$

$$= -(u_1 u_2 - v_1 v_2)$$

$$= -f\left(\begin{pmatrix}u_1\\u_2\end{pmatrix}, \begin{pmatrix}v_1\\v_2\end{pmatrix}\right)$$

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Dated:

- A permutation of set $S = \{1, \ldots, k\}$ is a bijective map $\sigma : S \to S$

$\{1, 2\}$
$\{2, 1\}$

matrix form:

$\sigma : S \to S$ where $S = \{1, 2, 3, 4, 5\}$

s.t $\sigma(1) = 2$, $\sigma(2) = 3$, $\sigma(3) = 4$, $\sigma(4) = 5$
$\sigma(5) = 1$

1) write it in matrix form:

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 1 \end{pmatrix} \begin{matrix} \to S \\ \to \text{image} \end{matrix}$$

2) write it in cyclic notation

$(1\ 2\ 3\ 4\ 5)$

Example 2: $(1, 3, 4)(2, 5)$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 4 & 1 & 2 \end{pmatrix}$$

→ Even permutation:

If a permutation can be written as product of even transpositions then permutation is even.

e.g:

$\leftarrow \sigma_1 = (1\ 2\ 3\ 4\ 5) = (12)(13)(14)(15)$

↓ even

$\sigma_2 = (1\ 2\ 3\ 4) = (12)(13)(14)$

↓ odd

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \to \text{always even.}$$

Identity Matrix

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Dated:

→ k- Linear Map:

Let $V^k = V \times V \times \cdots \times V$ (k copies of V)

$(v_1, v_2, \ldots, v_k)$

then $f: V^k \to \mathbb{R}$ is k- linear if it is linear in each argument.
The set of k- linear map is denoted by L_k & called k- tensor on V

Note:

$L_2 \to$ 2- Linear i.e: bilinear

$L_1 \to$ 1- Linear i.e: linear

- A k- tensor is symmetric if

$$f(v_1, \ldots, v_2) = \sigma f(v_1, \ldots, v_k)$$
$$= f(v_{\sigma(1)}, \ldots, v_{\sigma(k)})$$

where $\sigma \in S_k$

↳ permutation

- A k- tensor is alternating if

$$f(v_1, v_2, \ldots, v_k) = \text{sign } \sigma f(v_{\sigma(1)}, \ldots, v_{\sigma(k)})$$

- in case of 2- tensor

$$f(v_1, v_2) = f(v_2, v_1)$$

- in case of 3- tensor

$$f(v_1, v_2, v_3) \overset{(2,3)}{=} -f(v_1, v_3, v_2) \overset{(31)}{=} -f(v_3, v_2, v_1) \overset{(12)}{=} -f(v_2, v_1, v_3)$$
$$\underset{\substack{(123)\\(12)(13)}}{=} +f(v_3, v_1, v_2) \underset{\substack{(123)\\(12)(13)}}{=} +f(v_2, v_3, v_1)$$

∴ minus sign (-) because odd no. of permutations
plus sign (+) because even no. of permutations

Dated:

$f(x) = f(-x)$

$$f(x) = \underbrace{\frac{1}{2}\left(f(x) + f(-x)\right)}_{\text{even}} + \underbrace{\frac{1}{2}\left(f(x) - f(-x)\right)}_{\text{odd}}$$

→ Symmetrizing k-tensors

$$Sf = \sum_{\sigma \in S_k} \sigma f$$

in case of 2-tensor

$$= f(v_1, v_2) + f(v_2, v_1)$$

→ Alternating k-tensors

$$Af = \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma)\sigma f \rightarrow \text{2-tensor } f(v_1, v_2) - f(v_2, v_1)$$

The set of all alternating k-tensors is denoted by A_k

Ex: $f: \mathbb{R}^2 \times \mathbb{R}^2 \longrightarrow \mathbb{R}$

1) $f(u,v) = uv = v \cdot u = f(v,u) \rightarrow$ symmetric

2) $f(u,v) = u \times v = -v \times u = -f(v,u) \rightarrow$ alternator

3) $f\left(\begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}\right) = \det\begin{pmatrix} | & | \\ u & v \\ | & | \end{pmatrix} = -u_1 v_2 - u_2 v_1$

$$= f\left(\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix}\right) = \det\begin{pmatrix} v_1 & u_1 \\ v_2 & u_2 \end{pmatrix}$$

$$= v_1 u_2 - u_1 v_2$$

$$= -(u_1 v_2 - v_1 u_2) = -f(u,v)$$

alternating

Dated:

→ Tensor product

Let $f \in L_k$ $g \in L_m$ on V then

$(f \otimes g)(v_1, v_2, \ldots, v_k, v_{k+1}, \ldots, v_{k+m})$

$= f(v_1, \ldots, v_k)\, g(v_{k+1}, \ldots, v_{k+m})$

Wedge Product

$$f \wedge g = \frac{1}{k!\,m!} A(f \otimes g)$$

Q: $f: \mathbb{R}^2 \to \mathbb{R}$ $g: \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$

$f\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = u_1 u_2$ $g(u, v) = u \cdot v$

$(f \otimes g)(u, v, w) = f\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} g\left(\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}, \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}\right)$

Q: Let $f \in L_1 = A_1$ $g \in L_1 = A_1$

$f: \mathbb{R}^2 \to \mathbb{R}$ $g: \mathbb{R}^2 \to \mathbb{R}$

by $f\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = u_1 + u_2$ $g\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = u_1 - u_2$

1) $f \otimes g$ $(f \otimes g)(u, v)$

2) $f \wedge g$ $= f(u)\, g(v)$

$= (u_1 + u_2)(v_1 - v_2)$

$$(f \wedge g)(u, v) = \frac{1}{1!\,1!}\left[f(u)g(v) - f(v)g(u)\right]$$

$$= (u_1 + u_2)(v_1 - v_2) - (v_1 + v_2)(u_1 - u_2)$$

Q: Let $f \in A_2$ $g \in A_1$, Find

A means set of all alternating

$(f \wedge g)(v_1, v_2, v_3)$

$$\frac{1}{2!\,1!}\left[f(v_1, v_2)\, g(v_3) - f(v_2, v_1)\, g(v_3)\right.$$

$$+ f(v_1, v_3)\, g(v_2) - f(v_3, v_1)\, g(v_2) + f(v_2, v_3)\, g(v_1) -$$

$$f(v_3, v_2)\, g(v_1)$$